Old Kappa-Conditioned vs Revised Kappa-Collapsed Sampler

Collaborator-Facing Diagnostic Comparison

MSSTNB simulation diagnostics

2026-06-25

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1 Executive summary

The revised kappa-collapsed sampler answers the collaborator's concern favorably. It preserves posterior predictive calibration and produces nearly identical regression and baseline-risk recovery to the original sampler in S1–S3 and S4B–S4E. Its main advantage is robustness in S4A, the sparse-count stress test: the original sampler has one baseline-analysis failure out of ten replicates, whereas the revised sampler has no baseline-analysis failures.

On the common-success S4A replicates, the revised sampler reduces baseline log-risk RMSE from 0.2816 to 0.2625 and log-lambda RMSE from 0.4940 to 0.4025, while PPC 95% coverage is essentially unchanged (0.9952 versus 0.9951).

2 Purpose

This report compares two implementations of the MSSTNB sampler:

1. **Original kappa-conditioned sampler.** The regression and spatial baseline update is performed conditional on sampled κ.
2. **Revised kappa-collapsed sampler.** The baseline regression and spatial update is based on the marginal negative-binomial likelihood, with κ sampled later for the dispersion and dynamic-risk components.

The collaborator's question is whether the revised sampler removes baseline distortion or numerical fragility without degrading predictive performance. We therefore report four families of diagnostics:

- regression recovery for β_0, β_1, and β_2;
- recovery of the fitted baseline log-risk $\eta_{tj} = \beta_0 + \beta_1 x_{1,tj} + \beta_2 x_{2,tj} + \phi_j$;
- recovery of the dynamic multiplier on the log scale, $\log \lambda_{tj}$;
- posterior predictive calibration and the diagnostic association between $\log(\kappa_{tj})$ and the fitted baseline log-risk.

All comparisons use the same formal MCMC settings: 20,000 iterations, 10,000 burn-in iterations, and thinning by 5. The formal runs are single-chain, so formal R-hat is not available without a separate multi-chain rerun.

3 Why common-success comparison is needed

The original sampler had one baseline-analysis failure in scenario S4A. Comparing all revised fits against only successful old fits would mix different replicate sets in S4A. Therefore, the main metric comparison below is a **common-success comparison**: for each scenario, the revised sampler is restricted to the same replicate set for which the old sampler was baseline-usable. Failure rates are reported separately.

Table 1: Baseline-analysis status used to define the common-success replicate set.

Scen.	Total	Common success	Old failures	Old failure rate	Revised non-OK
S1	20	20	0	0.0%	0
S2	20	20	0	0.0%	0
S3	20	20	0	0.0%	0
S4A	10	9	1	10.0%	0
S4B	10	10	0	0.0%	0
S4C	10	10	0	0.0%	0
S4D	10	10	0	0.0%	0
S4E	10	10	0	0.0%	0

The revised sampler completed all 110 fits used in this diagnostic set without baseline-analysis failures. The original sampler was baseline-usable in all scenarios except S4A, where it had 1 failure out of 10 replicates, corresponding to a failure rate of 10.0%.

4 Regression recovery

Table 2: Regression RMSE on common-success replicates. Delta is revised minus old.

Scenario	Parameter	Old	Revised	Delta
S1	β_0	0.0124	0.0125	0.0001
S1	β_1	0.0127	0.0125	-0.0001
S1	β_2	0.0134	0.0137	0.0002
S2	β_0	0.0122	0.0122	0.0001
S2	β_1	0.0148	0.0148	0.0000
S2	β_2	0.0125	0.0124	-0.0001
S3	β_0	0.0124	0.0123	-0.0001
S3	β_1	0.0150	0.0150	0.0001
S3	β_2	0.0120	0.0125	0.0005
S4A	β_0	0.1266	0.1180	-0.0085
S4A	β_1	0.0435	0.0434	-0.0001
S4A	β_2	0.0384	0.0377	-0.0008
S4B	β_0	0.0588	0.0589	0.0001
S4B	β_1	0.0251	0.0250	-0.0001
S4B	β_2	0.0226	0.0226	0.0000
S4C	β_0	0.0263	0.0266	0.0003
S4C	β_1	0.0277	0.0273	-0.0004
S4C	β_2	0.0258	0.0253	-0.0005
S4D	β_0	0.0255	0.0259	0.0003
S4D	β_1	0.0303	0.0299	-0.0004
S4D	β_2	0.0258	0.0258	0.0000
S4E	β_0	0.0116	0.0115	-0.0001
S4E	β_1	0.0149	0.0143	-0.0006
S4E	β_2	0.0138	0.0149	0.0012

Across S1–S3 and S4B–S4E, the regression RMSE values are nearly identical between the two samplers. S4A is the only scenario in which stability differs: on the common-success replicates, the revised sampler has slightly smaller β_0 RMSE than the old sampler (0.1180 versus 0.1266).

5 Baseline-risk and dynamic-risk recovery

Table 3: Baseline log-risk and dynamic log-lambda recovery on common-success replicates. Delta is revised minus old.

Scen.	Base RMSE old	Base RMSE rev.	Delta base	log-lambda old	log-lambda rev.	Delta log-lambda
S1	0.0309	0.0312	0.0003	0.0000	0.0000	0.0000
S2	0.0299	0.0298	-0.0002	0.0469	0.0469	0.0000
S3	0.0301	0.0300	-0.0001	0.0489	0.0477	-0.0012

Scen.	Base RMSE old	Base RMSE rev.	Delta base	log-lambda old	log-lambda rev.	Delta log-lambda
S4A	0.2816	0.2625	-0.0191	0.4940	0.4025	-0.0916
S4B	0.1136	0.1135	-0.0001	0.2088	0.2094	0.0005
S4C	0.0586	0.0578	-0.0008	0.0740	0.0745	0.0006
S4D	0.0568	0.0566	-0.0002	0.0537	0.0535	-0.0002
S4E	0.0322	0.0322	-0.0001	0.0561	0.0561	0.0000

The baseline log-risk recovery is essentially unchanged outside S4A. Across S1–S3 and S4B–S4E, the largest absolute change in baseline log-risk RMSE is only 0.0008. In S4A, the revised sampler improves the common-success baseline log-risk RMSE from 0.2816 to 0.2625. It also improves the log-lambda RMSE from 0.4940 to 0.4025.

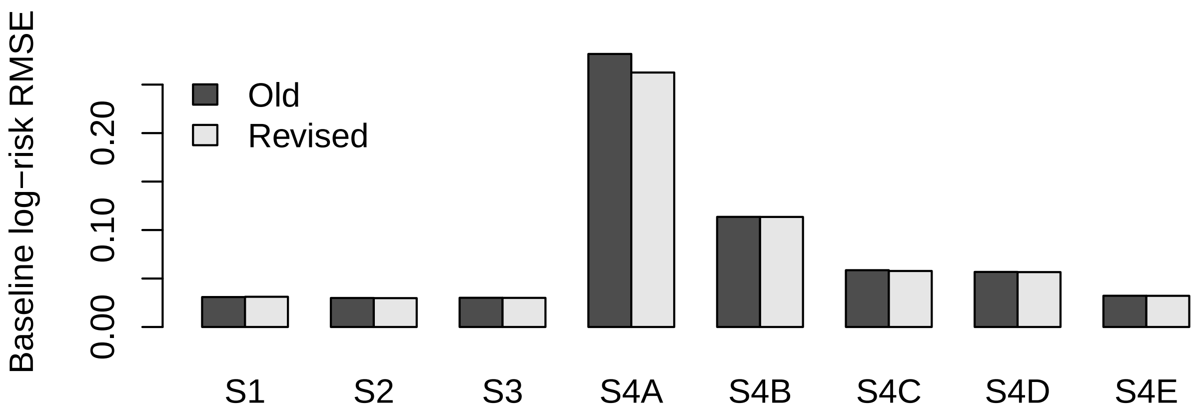

Figure 1: Baseline log-risk RMSE by scenario on common-success replicates.

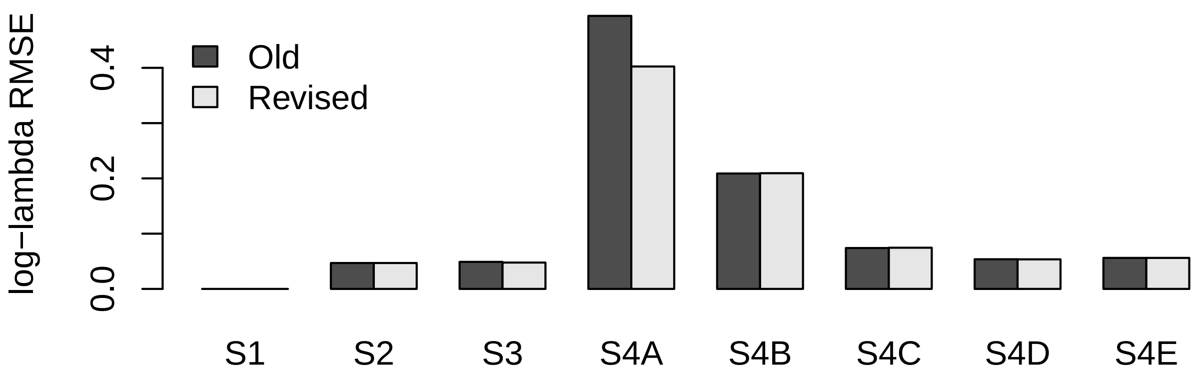

Figure 2: Log-lambda RMSE by scenario on common-success replicates.

6 Posterior predictive calibration

Table 4: Posterior predictive cell-level 95% coverage on common-success replicates. Delta is revised minus old.

Scenario	PPC 95% old	PPC 95% revised	Delta PPC 95%
S1	0.9596	0.9584	-0.0012
S2	0.9638	0.9634	-0.0003
S3	0.9639	0.9634	-0.0005
S4A	0.9952	0.9951	-0.0001
S4B	0.9874	0.9870	-0.0004
S4C	0.9589	0.9593	0.0004
S4D	0.9770	0.9785	0.0015
S4E	0.9649	0.9637	-0.0012

Posterior predictive calibration is not degraded by the revised sampler. The largest absolute change in 95% posterior predictive cell coverage is only 0.0015. In S4A, predictive coverage is essentially unchanged: old = 0.9952, revised = 0.9951.

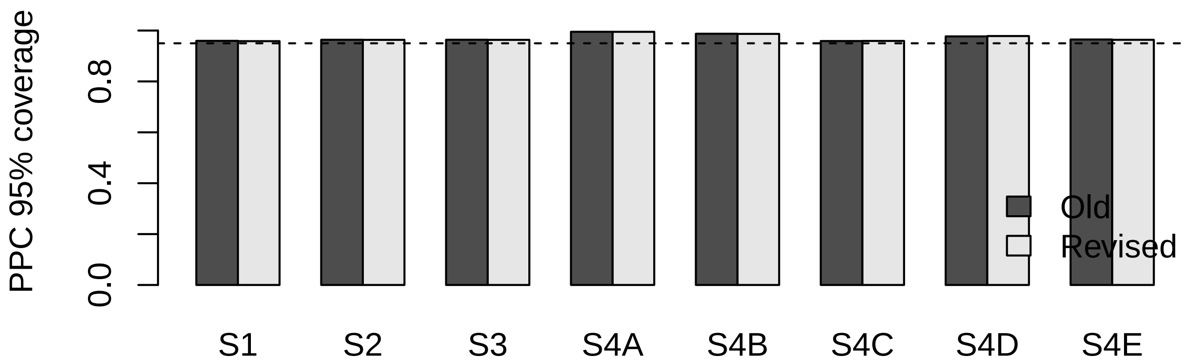

Figure 3: Posterior predictive 95% coverage by scenario on common-success replicates.

7 Diagnostic association between $\log(\kappa)$ and baseline log-risk

Table 5: Mean absolute draw-wise correlation between log-kappa and baseline log-risk. Delta is revised minus old.

Scenario	Abs cor old	Abs cor revised	Delta abs cor
S1	0.0269	NA	NA
S2	0.0270	0.0276	0.0006
S3	0.0273	0.0275	0.0002
S4A	0.0463	0.0450	-0.0013
S4B	0.0422	0.0417	-0.0005
S4C	0.0317	0.0308	-0.0009
S4D	0.0501	0.0508	0.0007
S4E	0.0271	0.0273	0.0002

The diagnostic association between $\log(\kappa)$ and the fitted baseline log-risk remains weak. Excluding S1, where the revised baseline-only fit does not store posterior κ samples and the association is not applicable, the largest absolute change in the mean absolute association is 0.0013. Thus, the revised sampler does not introduce stronger coupling between κ and the baseline log-risk.

8 Interpretation flags

Table 6: Automatic interpretation flags. Material baseline RMSE change is defined here as absolute delta greater than 0.01; material PPC drop is defined as PPC 95% coverage delta less than -0.01.

Scen.	Old fail rate	Base RMSE delta	log-lambda delta	PPC95 delta	Abs cor delta	Base material	PPC material drop	Old failure
S1	0.0%	0.0003	0.0000	-0.0012	NA	FALSE	FALSE	FALSE
S2	0.0%	-0.0002	0.0000	-0.0003	0.0006	FALSE	FALSE	FALSE
S3	0.0%	-0.0001	-0.0012	-0.0005	0.0002	FALSE	FALSE	FALSE
S4A	10.0%	-0.0191	-0.0916	-0.0001	-0.0013	TRUE	FALSE	TRUE
S4B	0.0%	-0.0001	0.0005	-0.0004	-0.0005	FALSE	FALSE	FALSE
S4C	0.0%	-0.0008	0.0006	0.0004	-0.0009	FALSE	FALSE	FALSE
S4D	0.0%	-0.0002	-0.0002	0.0015	0.0007	FALSE	FALSE	FALSE
S4E	0.0%	-0.0001	0.0000	-0.0012	0.0002	FALSE	FALSE	FALSE

Only S4A is flagged for a material baseline RMSE change and for old-sampler failure. This is expected because S4A is the sparse-count stress test. Importantly, S4A is not flagged for predictive degradation: the revised sampler preserves posterior predictive calibration while reducing baseline and log-lambda RMSE on the common-success replicate set.

9 Answer to the collaborator's question

The revised kappa-collapsed sampler does **not** degrade predictive performance. On common-success replicates, posterior predictive 95% coverage is virtually unchanged across all scenarios. The largest absolute change in PPC 95% coverage is 0.0015, and no scenario shows a material decrease.

The revised sampler also does **not** introduce a stronger diagnostic association between $\log(\kappa)$ and the fitted baseline log-risk. Outside S1, where the revised baseline-only fit does not store κ samples, the association remains weak and changes only negligibly.

The main practical advantage of the revised sampler is robustness under sparse counts. The original kappa-conditioned sampler has one baseline-analysis failure in S4A, whereas the revised sampler has no baseline-analysis failures in any scenario. On the common-success S4A replicates, the revised sampler reduces baseline log-risk RMSE from 0.2816 to 0.2625 and log-lambda RMSE from 0.4940 to 0.4025, while keeping PPC 95% coverage essentially unchanged.

Overall, the revised sampler provides a more stable and target-aligned implementation. Among successful fits, it agrees closely with the original sampler in the non-stress and non-sparse scenarios; under sparse counts, it avoids an old-sampler failure and improves latent baseline/dynamic-risk recovery without sacrificing posterior predictive calibration.

10 Files used

This report uses files generated by the diagnostic workflow. The source files are stored under `analysis_collab_diagnostics/old_vs_revised_compare/` and `analysis_collab_diagnostics/old_sampler_full_metrics/`.

Table 7: Input files used by this report.

Role	File
Common-success compact table	COMPARE4 compact table
Interpretation flags	COMPARE4 interpretation flags
Old common-success metrics	COMPARE4 old scenario metrics
Revised common-success metrics	COMPARE4 revised scenario metrics
Old failure/warning summary	OLD7 old failure summary
Revised failure/status summary	COMPARE3 revised status summary